

Vision LLM & AR Rendering: A Comprehensive Research Deep-Dive

Executive Summary

This report synthesizes the current state of Vision Language Models (VLMs) and AR rendering technologies as of 2025–2026, with a focus on building an integrated system that combines multimodal AI with real-time 3D visualization. The convergence of these fields is creating unprecedented opportunities for applications ranging from property intelligence to interactive AR experiences.

Part 1: Vision Language Models — The Foundation

1.1 The State of VLMs in 2025–2026

Vision Language Models have evolved from niche research artifacts to production-ready systems capable of sophisticated visual reasoning. By 2025, VLM research accounted for approximately **40% of all CVPR and ICLR abstracts**, up from just 16% in 2023. This explosive growth reflects the technology's maturation and expanding practical applications.

Key Architectural Trends:

Unified Understanding and Generation. A persistent challenge in VLMs has been the dichotomy between understanding and generation—treated as separate problems requiring fragmented architectures. **SenseNova-U1**, introduced in May 2026, addresses this with its NEO-unify architecture, treating understanding and generation as synergistic views of a single underlying process. The model comes in two variants (8B dense and 30B-A3B MoE) and demonstrates strong performance across text understanding, vision-language perception, knowledge reasoning, and spatial intelligence. Notably, it also shows promise in vision-language-action (VLA) and world model scenarios.

Interpretable Visual Representations. **Cascaded Sparse Autoencoders** (June 2026) provide a scalable way to decompose dense model activations into sparse, interpretable features, addressing the "black box" nature of multimodal LLMs.

Efficient Architectures. **Firebolt-VL** replaces the Transformer-based decoder with an efficient Liquid Foundation Model, enabling deployment in resource-constrained scenarios such as personal assistants and smart cameras.

1.2 Major VLM Families (2025–2026)

Model Family	Key Features	Best Use Cases
SenseNova-U1	Unified understanding + generation, 8B-30B params, MoE variant	General multimodal tasks, VLA, world modeling
InternVL3 (OpenGVLab, 2025)	Third-generation vision-language model, strong visual reasoning	Visual QA, image understanding
DeepSeek-VL2	Advanced visual reasoning capabilities	Complex visual reasoning
AR-Omni (GitHub)	Single-decoder, single-token-stream any-to-any generation (text, images, speech)	Multimodal generation
Dream-VL	Diffusion language model backbone, superior visual planning	Visual planning tasks
MiniCPM-o 4.5	Vision, speech, full-duplex multimodal live streaming	Real-time multimodal interaction

1.3 VLMs for 3D Scene Understanding

This is the most critical area for AR applications. Several breakthrough approaches emerged in 2025–2026:

PAR3D (June 2026). A unified part-aware 3D-MLLM framework that enables models to understand, reason about, and ground both objects and their **parts** in 3D scenes. Unlike object-centric approaches, PAR3D models fine-grained part structures essential for embodied interaction. It introduces ScenePart, a synthetic 3D scene dataset with part-level annotations.

View-on-Graph (December 2025). A zero-shot 3D visual grounding method that organizes scenes into multi-modal, multi-layer scene graphs. The VLM operates as an active agent that selectively accesses necessary cues while traversing the scene—a paradigm shift from processing entire cluttered visual inputs. This yields state-of-the-art zero-shot performance with interpretable step-by-step traces.

Loc3R-VLM (March 2026). Equips 2D Vision-Language Models with advanced 3D understanding capabilities from monocular video input.

LLaVA³ (March 2026). Represents 3D scenes "like a Cubist painter" to boost 3D scene understanding of VLMs, addressing the limited availability of 3D training data.

3DCity-LLM (March 2026). A unified framework for 3D city-scale vision-language perception using coarse-to-fine feature encoding.

3DGraphLLM (August 2025). Combines semantic graphs and LLMs for 3D scene understanding, explicitly incorporating semantic relationships.

OV-SCAN (2025). Open-vocabulary 3D object detection connecting traditional detectors with VLMs to discover novel objects.

LocateAnything3D (November 2025). A VLM-native recipe treating 3D detection as a next-token prediction problem with Chain-of-Sight reasoning.

1.4 VLM Benchmarks & Evaluation

The CVPR 2025 workshop survey provides a comprehensive overview of VLMs developed up to 2025, covering architecture transitions, alignment methods, benchmarks, and challenges including hallucination, alignment, fairness, and safety.

Key Challenges Identified:

- **Hallucination:** VLMs sometimes generate plausible but incorrect visual descriptions
 - **Alignment:** Ensuring model outputs align with human expectations
 - **Fairness & Safety:** Bias in training data and potential misuse
 - **Spatial Reasoning:** VLMs still struggle with viewpoint-aware reasoning
-

Part 2: AR Rendering Technologies

2.1 The Rendering Technology Landscape

Real-time 3D reconstruction and rendering are essential for immersive AR/VR experiences but remain challenging with traditional photogrammetry and mesh-based pipelines due to high latency and limited scalability.

The Three Major Approaches:

Approach	Strengths	Weaknesses
NeRF (Neural Radiance Fields)	Photorealistic quality, continuous scene representation	Long training times, slow rendering, computationally intensive
3D Gaussian Splatting (3DGS)	Fast training, real-time rendering, photorealistic quality	Heavy preprocessing, memory usage
Traditional Meshes	Mature tooling, hardware acceleration	High reconstruction cost, limited visual realism

2.2 3D Gaussian Splatting — The Current State of the Art

3D Gaussian Splatting (Kerbl et al., 2023) models scenes as sets of view-dependent Gaussians splatted directly onto the image plane using a fast, differentiable rasterization process. This enables faster training and real-time rendering while maintaining photorealistic quality.

Recent Advances (2025–2026):

Capture and Interact (2025). An end-to-end pipeline leveraging 3DGS for rapid acquisition and interactive rendering using a mobile device, cloud processing, and Unity. Users scan objects with smartphone video, upload for automated 3D reconstruction, and visualize interactively in Unity at **~150 fps**. Processing takes approximately **10 minutes on GPU**.

3DGauCIM (July 2025). Accelerates static/dynamic 3DGS for high frame rate real-time edge rendering, addressing the challenges of deploying dynamic 3DGS on edge devices.

IRIS (October 2025). An energy-efficient spatial computing SoC for real-time interactive rendering and modeling with surface-aware 3DGS.

SplatLoc (2025). A 3DGS-based visual localization method for AR, leveraging Gaussian primitives as scene representation.

TC3DGS (2025). Temporally Compressed 3D Gaussian Splatting for compressing dynamic 3D Gaussian representations.

SAGOnline (August 2025). Enables efficient and consistent 3D segmentation of Gaussian splats, paving the way for interactive 3D understanding in AR/VR.

EGGS (NeurIPS 2025). Exchangeable Gaussian Splatting—a hybrid representation integrating 2D and 3D Gaussians to balance appearance and geometry.

2.3 Single-Image to 3D Generation

AR-1-to-3 (ICCV 2025). A novel paradigm that progressively generates target views in an autoregressive manner. Rather than producing multiple discrete views simultaneously, it starts from views closer to the input view and uses them as contextual information to generate farther views. This maintains **cross-view consistency** even with significantly different camera poses.

SAM 3D (2025). Transforms single 2D images into detailed 3D models of humans, objects, and complete scenes in seconds through three specialized tools: Body, Objects, and Align.

MMPart (2025). Generates part-aware 3D models from a single image by first using a VLM to generate prompts based on the input image and user descriptions.

DreamArt (2025). Generates high-fidelity, interactable articulated assets from single-view images using a three-stage pipeline: image-to-3D generation, mask-prompted 3D segmentation, and reconstruction.

2.4 NeRF for AR

Cloud-Edge NeRF (July 2025). A cloud-edge collaborative framework integrating Visual SLAM with NeRF for real-time virtual-real fusion rendering in AR.

Adaptive Sparse NeRF (December 2025). Combined with efficient spatial indexing and a hybrid rendering pipeline to improve real-time 3D reconstruction quality.

Scale-NeRF. Demonstrates excellent dynamic reconstruction capabilities with real-time rendering.

Part 3: Mobile AR Platforms & SDKs

3.1 Apple ARKit + RealityKit

Apple's AR ecosystem provides a powerful toolkit for spatial computing:

- **ARKit** provides detailed contextual information about the real-world environment
- **RealityKit** is the primary way to display 3D content across iOS, iPadOS, macOS, tvOS, and visionOS
- **visionOS 26 (WWDC25)** features enhanced Volumetric API and improved integration between SwiftUI, RealityKit, and ARKit
- **RoomPlan APIs** enable real-time spatial mapping and 3D reconstruction of interior environments

Optimization: 3D models for ARKit require careful attention to polygon count, file size, texture resolution, and draw calls.

3.2 Google ARCore

Google's AR platform offers robust 3D reconstruction capabilities:

- **ArMesh** provides a triangulated mesh representing a surface reconstruction of the scene
 - **Depth API** uses a depth-from-movement algorithm to create depth images, offering a 3D view of the world
 - **Geospatial Creator** provides access to the same 3D map source used by Google Earth
-

Part 4: Integration — VLM + AR

4.1 Generative AI for AR

Generative AI for Context-Aware 3D Object Creation (February 2025). A system functioning as a designer assistant in AR environments, leveraging VLMs like LLaVA and advanced text-to-3D generative models. Users can capture images of their surroundings with an AR headset to generate context-aware 3D objects.

AI3D Co-Create (2025). A framework supporting both real-time and "rendered" 3D multimodal controlled inputs, enabling image-to-image-to-3D workflows.

Transcending Dimensions (April 2025). A cohesive system integrating generative AI models like Shap-E to transform 2D images into 3D representations in AR environments.

4.2 Google Gemini for AR

Google's Gemini API is increasingly used for AR applications:

- **Gemini 3 Flash** analyzes images and returns structured JSON with bounding boxes, labels, and collision prompts—ideal for AR
- **Gemini 1.5 Pro** provides vision capabilities for 3D object tracking and AI assistance
- **Gemini 2.5 Flash** powers creative prompts into interactive XR experiences, generating polished 3D textures
- **GenCAD** uses Gemini's vision capability to capture screenshots of rendered 3D models and verify output matches user requests

4.3 3D Scene Graphs for VLM-AR Integration

A critical insight from recent research: **3D scene graphs** provide the structured representation that bridges VLMs and AR systems.

View-on-Graph demonstrates that organizing scenes into multi-modal, multi-layer scene graphs allows VLMs to operate as active agents that selectively access necessary cues. This reduces reasoning difficulty and produces transparent, interpretable traces.

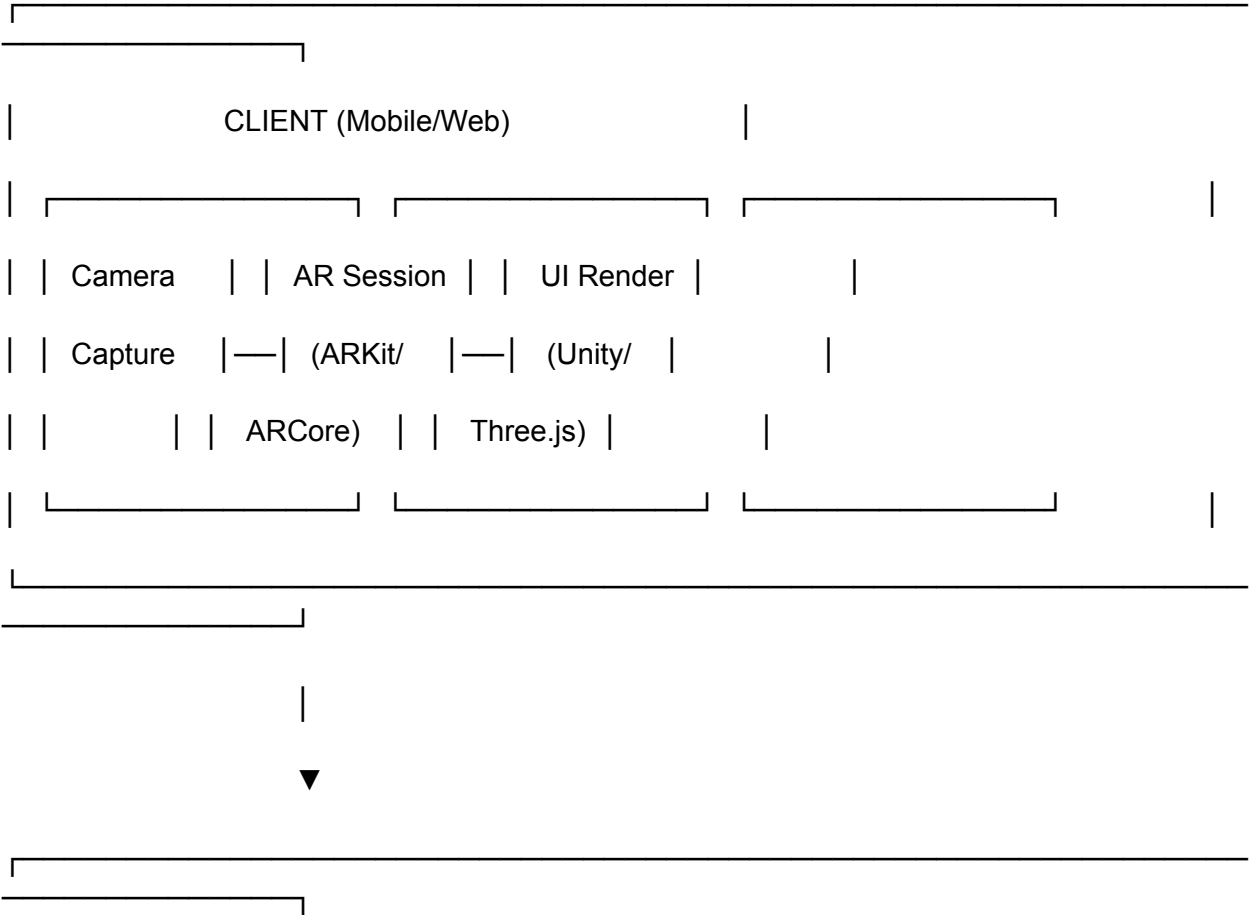
Dynamic Open-Vocabulary 3D Scene Graphs (DovSG, February 2025). Enables long-term language-guided mobile manipulation by leveraging dynamic open-vocabulary 3D scene graphs.

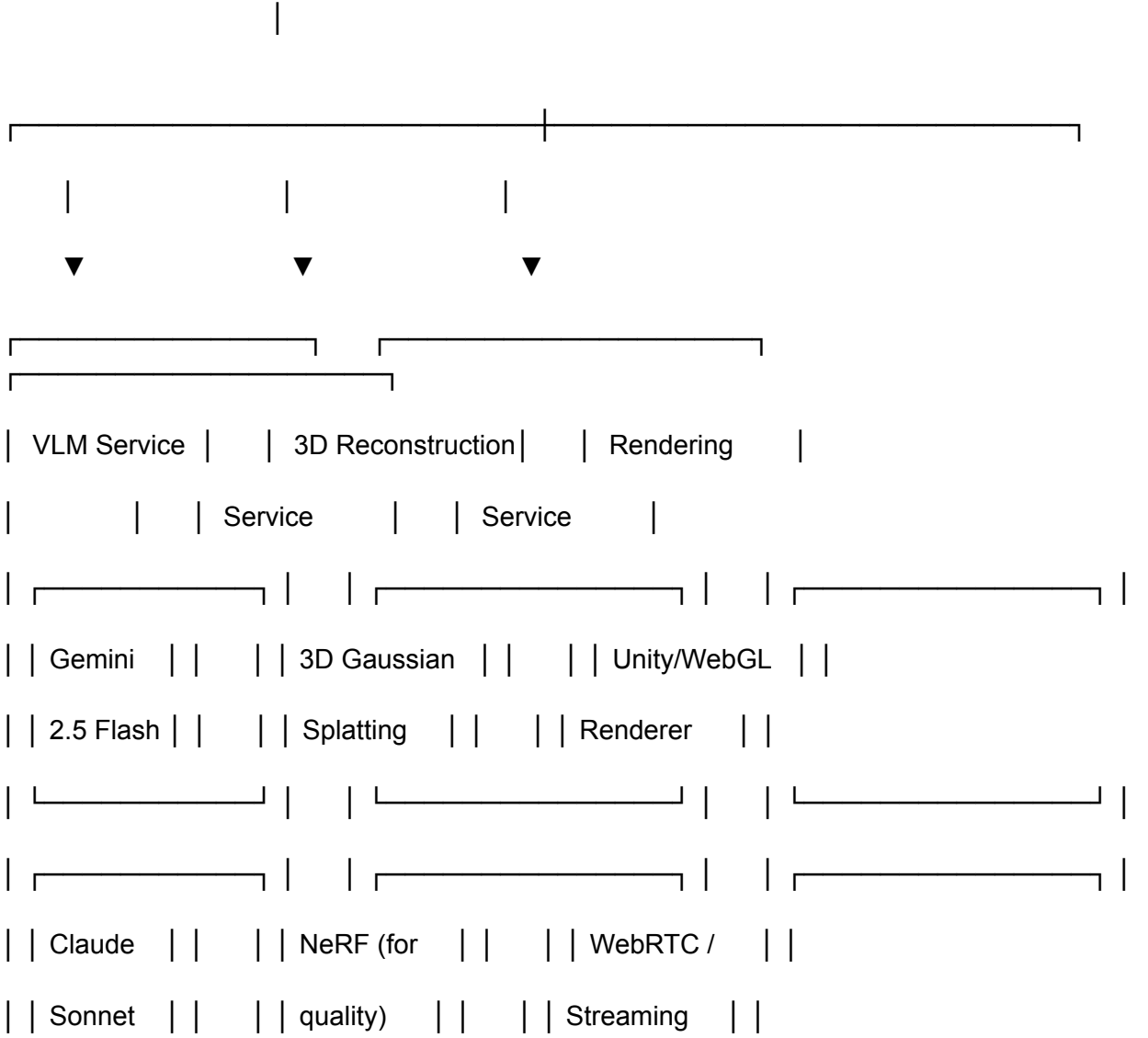
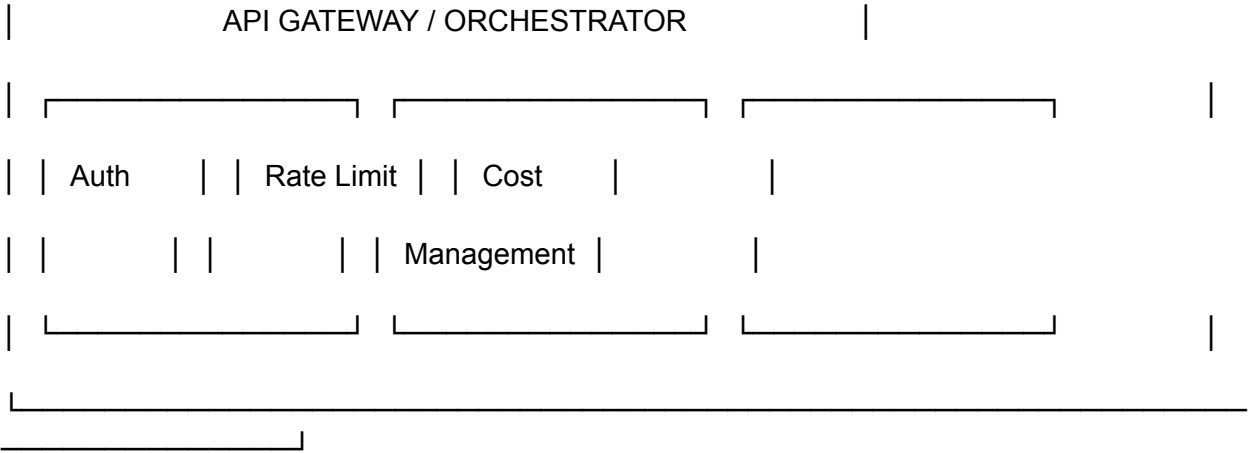
LSVG (August 2025). Constructs language-guided scene graphs with 2D-assisted multi-modal encoding for 3D visual grounding.

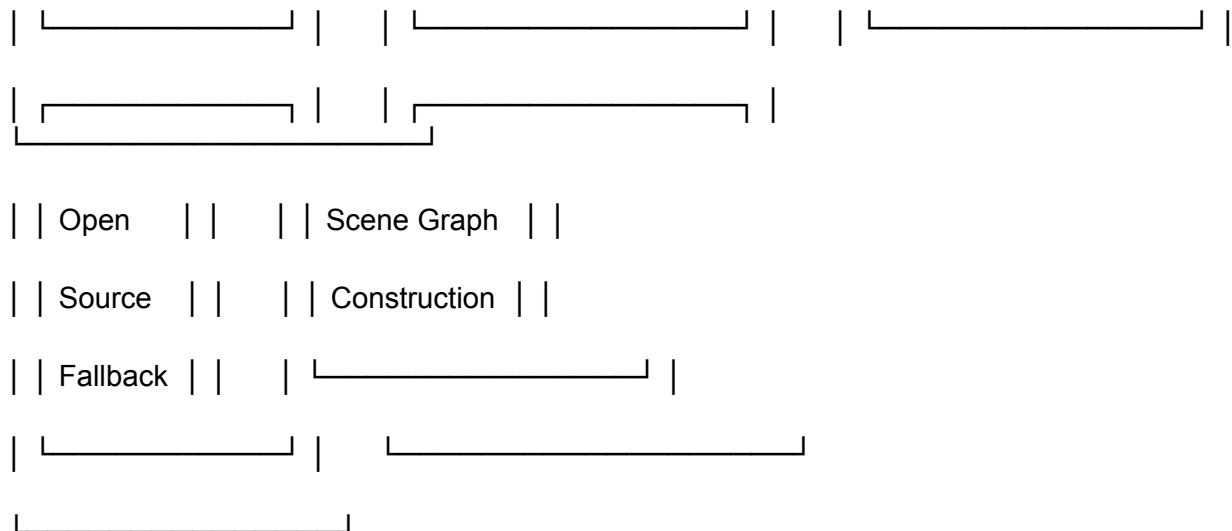
Part 5: Building a Working VLM + AR System

5.1 Recommended Architecture

Based on the research, here is a recommended system architecture:







5.2 Component Selection

Vision Language Model:

Component	Recommendation	Rationale
Primary VLM	Google Gemini 2.5 Flash	Fast inference, vision capabilities, structured JSON output
Fallback	Claude Sonnet or DeepSeek-VL2	Resilience against API outages
Open Source	InternVL3 or Dream-VL	For on-premise/private deployments
3D-Specialized	PAR3D or Loc3R-VLM	For 3D scene understanding tasks

AR Rendering:

Component	Recommendation	Rationale
Mobile AR	ARKit (iOS) / ARCore (Android)	Platform-native, hardware acceleration
3D Reconstruction	3D Gaussian Splatting	Real-time performance, photorealistic quality

Component	Recommendation	Rationale
Fallback	NeRF	Higher quality for non-real-time scenarios
Scene Graph	View-on-Graph approach	Structured representation for VLM reasoning
Unity Integration	3DGS-to-Unity pipeline	~150 fps rendering

5.3 Implementation Phases

Phase 1: Core VLM Integration

- Set up Gemini API with structured JSON output
- Implement multi-provider fallback (Gemini → Claude → Open Source)
- Build prompt engineering pipeline for property/object analysis
- Implement response caching for cost optimization

Phase 2: 3D Reconstruction Pipeline

- Implement 3D Gaussian Splatting for object capture
- Build mobile capture → cloud processing → local rendering pipeline
- Optimize for ~10-minute processing time
- Integrate with Unity for real-time rendering

Phase 3: Scene Understanding

- Implement 3D scene graph construction
- Integrate VLM for scene reasoning and object grounding
- Enable language-guided interaction (e.g., "show me the damaged area")
- Implement part-aware segmentation using PAR3D approach

Phase 4: AR Experience

- Build ARKit/ARCore session management
- Implement real-time virtual-real fusion rendering
- Add gesture/voice interaction
- Deploy to mobile platforms

5.4 Cost Optimization

Based on the research, implement these strategies:

Strategy	Expected Savings
Response caching (hash inputs, TTL 24h)	30-40%
Request batching (group similar analyses)	20-25%
Model tiering (Flash for 90% of requests)	40-50%
Fine-tuned local models (repetitive tasks)	60-70%
Prompt compression	10-15%

5.5 Key Technical Decisions

1. Scene Representation: 3DGS vs. NeRF

- **Use 3DGS** for real-time AR applications where speed is critical
- **Use NeRF** for high-quality offline rendering or when processing time is less constrained
- Consider **hybrid approaches** like EGGS for balanced geometry-appearance

2. VLM Model Selection

- **Gemini 2.5 Flash** is recommended as primary due to vision capabilities and structured output
- **Open-source alternatives** (InternVL3, Dream-VL) for privacy-sensitive deployments
- **PAR3D** or **Loc3R-VLM** for 3D-specific understanding tasks

3. Scene Graph Strategy

- Adopt the **View-on-Graph paradigm** for zero-shot 3D visual grounding
- Build **multi-modal, multi-layer scene graphs** that enable selective VLM reasoning
- Use **Language Sensor Models** for calibrated spatial belief estimation

4. AR Platform

- **Apple:** ARKit + RealityKit for iOS/visionOS
 - **Google:** ARCore + Geospatial API for Android
 - **Cross-platform:** Unity with 3DGS integration for maximum reach
-

Part 6: Research Gaps & Future Directions

6.1 Current Limitations

- 1. **Real-time 3D reconstruction remains compute-intensive.** While 3DGS has reduced training time, mobile inference still challenges edge devices.
- 2. **VLM 3D understanding is nascent.** Most VLMs are trained on 2D data; 3D understanding requires new architectures and datasets.
- 3. **Scene graph construction is not fully automated.** Manual intervention is often required for complex scenes.
- 4. **Latency.** End-to-end latency from capture to AR rendering remains a challenge.
- 5. **Hallucination in VLMs.** Incorrect visual descriptions can break AR experiences.

6.2 Future Research Directions

- 1. **Unified understanding-generation models** (e.g., SenseNova-U1) that can both perceive and create
- 2. **Part-aware 3D understanding** for fine-grained interaction
- 3. **Zero-shot 3D grounding** for open-vocabulary AR scenarios
- 4. **Energy-efficient edge rendering** for mobile AR
- 5. **Dynamic scene reconstruction** for moving objects in AR
- 6. **VLM-native 3D detection** treating 3D detection as next-token prediction

Part 7: Getting Started — Quick Reference

7.1 Recommended Open Source Resources

Resource	Description	Link
PAR3D	Part-aware 3D-MLLM	github.com/atrovast/PAR3D
View-on-Graph	Zero-shot 3D visual grounding	arxiv.org/abs/2512.09215

Resource	Description	Link
AR-Omni	Any-to-any generation	github.com/ModalityDance/AR-Omni
Capture and Interact	3DGS + Unity pipeline	ar5iv.labs.arxiv.org/html/2510.06802
InternVL3	Open-source VLM	github.com/OpenGVLab/InternVL
Dream-VL	Diffusion-based VLM	arxiv.org/abs/2601.02468

7.2 Key APIs

API	Purpose	Documentation
Gemini API	Vision + structured JSON + image generation	ai.google.dev
ARKit	iOS AR + 3D reconstruction	developer.apple.com/arkit
ARCore	Android AR + 3D reconstruction	developers.google.com/ar
Unity 3DGS	Real-time rendering	github.com/Unity-Technologies

7.3 Immediate Action Items

1. **Set up Gemini API** with structured output for property/object analysis
2. **Implement 3DGS capture pipeline** using smartphone video
3. **Build Unity renderer** for ~150 fps visualization
4. **Integrate scene graph** for VLM reasoning
5. **Deploy AR session** using ARKit/ARCore for real-world overlay

Conclusion

The convergence of Vision Language Models and AR rendering technologies has reached a critical inflection point. Modern VLMs like Gemini 2.5 Flash, SenseNova-U1, and PAR3D can now understand 3D scenes with unprecedented accuracy, while 3D Gaussian Splatting enables real-time photorealistic rendering at ~150 fps on consumer hardware.

The key to building a working system lies in the **scene graph abstraction**—structured representations that bridge VLM reasoning with AR rendering. By organizing 3D scenes into multi-modal, multi-layer graphs, VLMs can selectively reason about objects and their relationships, while AR systems can render them in real-time.

The path forward is clear: integrate a primary VLM (Gemini 2.5 Flash) with a 3DGS-based rendering pipeline (Unity), orchestrated through scene graphs, and deployed on mobile AR platforms (ARKit/ARCore). This combination delivers the intelligence to understand scenes and the performance to render them interactively.

Atlas does not sell software. Atlas sells certainty—and certainty wins contracts. In the context of VLM + AR, certainty means knowing what you're looking at and showing it convincingly, in real-time, in the real world.

EagleView & ServiceTitan — Complete Technical & Feature Blueprint

Authored from the unified perspective of a Product-Business-Technical-Market Architect.

This document provides a rigorous, tech-spec-style dissection of two distinct platforms: **EagleView** (aerial property intelligence) and **ServiceTitan** (home-service business operating system). Each section reads as an architectural blueprint covering full feature taxonomy, underlying domain models, integration surfaces, infrastructure topology, user-journey mapping, and business-metric anchoring.

PART 1: EAGLEVIEW — PROPERTY INTELLIGENCE PLATFORM

1. Executive Summary & Market Positioning

EagleView is the market leader in geospatial property analytics, owning a proprietary capture fleet (fixed-wing aircraft, drones, satellites) and a machine-learning processing pipeline that delivers structural measurements, damage assessment, and 3-D property models to four verticals: **Insurance**, **Construction**, **Solar**, and **Government**. The platform fuses oblique and orthogonal aerial imagery with photogrammetry and computer vision to generate sub-inch-accurate roof reports, wall measurements, and change-detection alerts. The product portfolio is sold as API-first data services and SaaS-based web/mobile workbenches.

Key Business KPIs

- Policy Lifecycle Time Reduction (insurance carriers)
 - Claim Accuracy Index (supplement reduction)
 - Solar Install Close Rate & Design Time
 - Construction Material Takeoff Precision ($\leq 2\%$ error)
 - Parcel Change-Detection Recall ($\geq 95\%$)
-

2. Product Ecosystem & Feature Taxonomy

2.1 Core Product Lines

1. **EagleView Assess** – Property measurement reports (roof, siding, gutter, total-station-grade).
2. **EagleView OnSite** – Interactive 3-D digital twin, pin-and-measure, annotation.
3. **EagleView Cloud API** – JSON/GeoTIFF integration endpoints.
4. **EagleView Solar** – Solar-ready roof shading analysis, panel layout, irradiance maps.
5. **EagleView Construction** – Construction trade tools with cut-list and material ordering integration.
6. **EagleView Detect** – Change detection over time for tax assessment and compliance.

2.2 Functional Blueprint (per vertical)

A. Insurance Claims Workflow

Personas: Desk Adjuster, Field Adjuster, Claims Manager, SIU Investigator.

- **Instant Roof Report:** slope, area, ridge/hip/rake lengths, eave/valley dimensions, pitch per facet, waste factor, underlayment quantities.
- **Damage Detection:** hail spatter polygons, missing shingles, wind-torn membrane, tarp detection via deep learning, classified by severity.
- **Historical Imagery Compare:** slider view of pre/post event imagery, time-travel to validate storm date.
- **Sketch Integration:** export to Xactimate, Symbility, Duck Creek, Guidewire in ESX/XML.
- **Automated Claim Triage:** rules engine scores claim complexity, auto-approves low-risk claims.
- **Fraud Flags:** roof condition mismatch with reported DOL, pre-existing damage marking.

B. Solar Design & Installation

Personas: Solar Sales Consultant, Design Engineer, Installer.

- **Solar-Ready Roof Map:** azimuth per facet, pitch, roof-mounted obstruction detection (vents, skylights), usable square footage.
- **Shading Analysis:** annual sun path with hourly irradiance, tree obstacle polygons, LiDAR-derived DSM.
- **PV Layout Generator:** panel placement optimizer constrained by fire setbacks, structural loading, NEC code; outputs stringing diagram.
- **Electrical & Structural Summary:** service panel location suggestion, point-load calculations.
- **API Export:** Aurora Solar, Solo, SunDAT, PVsyst format.

C. Construction & Roofing

Personas: Roofing Contractor, Estimator, Material Supplier.

- **Detailed Material Takeoff:** shingle bundles, starter/ridge cap, underlayment squares, drip edge linear feet, valley metal, pipe boots.
- **Waste Calculator:** per-facet waste factor based on pitch and shape complexity.
- **3-D Visualized Estimate:** contractor-facing OnSite project with branded report.
- **Order Integration:** direct link to ABC Supply, Beacon, SRS Distribution with material list push.
- **Job-Progress Integration:** sync measurements with job phases.

D. Government & Tax Assessment

- **Parcel Change Detection:** building footprint delta, impervious surface change, pool addition, ADU detection—triggers reassessment.
- **GIS Integration:** WMS/WMTS layers for ArcGIS, QGIS; parcel fabric alignment.
- **Canopy Height & Landcover Classification:** supporting stormwater billing and urban planning.

3. Platform Technical Architecture (C4 Model)

Level 1 — System Context

- **Image Capture Fleet** → Ingest Service (RAW images + GNSS/IMU metadata)
- **Photogrammetry Pipeline** (aerial triangulation, point cloud, mesh, orthorectification)
- **ML Inference Engine** (facet segmentation, damage classifier, change detector)
- **EagleView Core API Layer** (REST, OAuth2, rate-limited)
- **Client Apps:** Web-Angular Assess UI, React-Native mobile OnSite, third-party integrations via API.
- **Data Export Handlers:** SFTP, webhooks, direct cloud bucket drop.

Level 2 — Containers

[Capture Scheduler] → [Flight/Sat Queue] → [Raw Image Store: S3]

→ [AT/Photogrammetry Service: GPU cluster] → [3-D Mesh Store]

→ [ML Segmentation Service: PyTorch/TensorRT] → [Property Graph DB: Neo4j/Cosmos]

→ [Report Generation: headless Chromium + LaTeX] → [CDN]

→ [API Gateway: Kong] → [Microservices: Property, User, Order, Billing]

→ [Data Lake: Snowflake] ← [Analytics Pipeline: dbt]

Level 3 — Key Component Design

- **Photogrammetry Engine:** Stateful pipeline using Agisoft Metashape / custom SfM; outputs dense point cloud → textured mesh. Each job is idempotent; correlation ID per property centroid.
- **Facet Segmentation Service:** Instance segmentation (Mask R-CNN) on oblique views, project to 3-D model. Stores facet polygons in PostGIS.
- **Measurement Engine:** Precise Euclidean & geodesic computations on the mesh; pitch angle, area with slope correction. Accuracy maintained to $\pm 2''$ on 95th percentile.
- **Change Detector:** Siamese network comparing orthoimage pairs; delta vectors classified as “structure,” “vegetation,” “debris.”
- **Report Renderer:** JSON template → headless Chrome renders interactive 3-D; PDF generated via LaTeX overlay with vector measurement lines.
- **Property Graph:** A property node with edges to imagery, reports, orders, change events; enables fast time-series traversal.

4. Data Model (Logical Entity-Relationship)

- **Property** (propertyId, centroid [lat/lon], address, parcelId, boundaryPolygon)
- **ImagerySet** (imageryId, propertyId, captureDate, source[fly|sat|drone], azimuth, elevation, s3URL)
- **RoofModel** (modelId, propertyId, version, meshURL, pointCloudURL)
- **RoofFacet** (facetId, modelId, geometry, pitch, area, azimuth, surfaceMaterial)
- **MeasurementReport** (reportId, propertyId, modelId, orderId, type, accuracyLevel, jsonData, pdfURL)
- **DamageEvent** (eventId, propertyId, stormDate, perilType, damagePolygons[])
- **SolarDesign** (designId, propertyId, panels[], shadingData, irradianceMapURL)
- **Order** (orderId, userId, propertyId, productType, status, createdAt, billingRef)
- **User/Organization** (tenantId, vertical, permissionRoles)

5. API & Integration Surface

EagleView Cloud API v3 (REST/JSON)

- **Endpoints:** `/properties`, `/reports`, `/orders`, `/models/{id}/facets`, `/solar/designs`
- **Webhook Events:** `report.completed`, `model.processing.error`, `order.status.changed`
- **File Delivery:** Presigned S3 URLs, automatic Dropbox/G-Drive export, SFTP push.
- **Partner Integration SDK:** Xactware XactAnalysis Bridge, Guidewire Edge, Duck Creek Anywhere.
- **Rate Limiting:** token bucket per clientId; burst 100 req/s, sustained 20 req/s.

Geospatial Services:

- **WMS/WMTS endpoints** for imagery tiles (EPSG:3857/4326)
 - **GeoJSON export** of facets, damage, solar panels.
 - **ArcGIS Pro add-in** and QGIS plugin.
-

6. User Experience & Interaction Patterns

- **Assess Web UI:** Map-centric dashboard, property search, layer toggles (aerial/oblique/solar), facet highlight on click, measurement pane, order wizard (3-step checkout).
 - **OnSite Interactive Viewer:** Three.js-based 3-D model; click-to-measure distance/area/pitch, pin annotations, time-slider for imagery comparison, AR mode on iPad (LiDAR-assisted on-site overlay).
 - **Mobile Capture:** Field adjuster app for guided photo capture (holds image orientation metadata for AI), on-device damage detection, offline queue-sync.
-

7. Infrastructure, Scalability & Cost Model

- **Cloud:** AWS (primary), Azure (government cloud).
- **Compute:** GPU-optimized EC2 (p3/p4) for photogrammetry and inference, auto-scaling via job queues (AWS Batch).
- **Storage:** S3 Standard for imagery (warm), Glacier for archive; DynamoDB for orders, RDS PostgreSQL + PostGIS for spatial, ElastiCache for session/semantic cache.
- **Data Pipeline:** Kinesis for ingestion events, SNS/SQS for job status, Step Functions orchestrate multi-stage processing.

- **Multi-Region:** Active-passive for report generation; imagery partitioned by capture region.
 - **Cost Efficiency:** Spot instances for batch processing, reserved instances for API tier. FinOps tag for cost-per-report telemetry; target gross margin $\geq 70\%$.
 - **SLIs/SLOs:** API latency p99 < 800 ms, Report generation < 45 min for standard, Uptime 99.9% (API), 99.5% (model generation).
-

8. Security & Compliance

- **Zero Trust Architecture:** mTLS for internal services, SPIFFE identity.
 - **RBAC:** organization-scoped tokens, fine-grained permissions (view imagery, order report, manage users).
 - **Data Privacy:** CCPA/GDPR opt-out on property imagery via blurring pipeline; data retention configurable per customer.
 - **Audit:** all access and order events logged to immutable bucket, feed SIEM.
 - **Image Integrity:** cryptographic hash on capture, tamper-evident chain for claims defense.
-

PART 2: SERVICE TITAN — HOME SERVICES OPERATING SYSTEM

1. Executive Summary & Market Positioning

ServiceTitan is a vertical SaaS platform purpose-built for residential and commercial home-service contractors (HVAC, plumbing, electrical, roofing, garage door, etc.). It combines CRM, field-service management, dispatch, invoicing, payments, marketing automation, reporting, and inventory management into a mobile-first, multi-tenant cloud system. It acts as the system-of-record for a contractor's entire business, driving operational efficiency, revenue growth, and customer lifetime value.

Business KPIs Linked to Platform

- Booked Revenue (daily/weekly)
- Technician Utilization Rate
- Average Ticket Value (ATV)
- Booking-to-Cash Cycle Time
- Membership Plan Attach Rate
- Marketing ROI (cost-per-lead by source)

2. Functional Module Blueprint

2.1 CRM & Customer 360

Personas: CSR, Dispatcher, Field Tech, General Manager.

- **Customer Record:** contact info, location(s) with property details, equipment inventory, service history, call recordings, tags, lead source, marketing consent.
- **Duplication Checker:** fuzzy match on phone/email during call-book to merge records.
- **Call-Booking Screen (TitanAdvisor):** intelligent call scripting; screen pop with caller ID, CSR sees prior visits, open estimates, membership status. One-click book with capacity management.
- **Job Management:** create/update job, assign to technician, status flow (scheduled → en route → on-site → completed → invoiced → paid).
- **Location Hierarchy:** parent/child location for property managers, multi-unit buildings.
- **Custom Fields & Tags:** user-defined fields on customer, location, job, invoice.

2.2 Dispatch & Scheduling

- **Dispatch Board:** drag-and-drop cards with tech capacity, skills, and travel time; color-coded by job status, priority, required parts.
- **Optimization Engine:** route optimization considering time windows, technician proximity, SLA; suggests re-order to maximize revenue density.
- **Capacity Management:** tech schedule blocks (PTO, training), overbooking alerts, arrival window SLAs.
- **GPS Integration:** real-time technician location on map; ETA push notifications to customers.
- **Arrival & Departure Check-Ins:** geofence-based automatic check-in; triggers “technician is on the way” SMS.

2.3 Mobile Field App (TitanPro)

Personas: Field Technician, Installer.

- **Job Overview:** address, notes, equipment, past jobs, customer alerts (dogs, gate codes).
- **Digital Forms:** configurable forms (safety, check-list, IAQ survey) with photo capture, signature, conditional logic.
- **Diagnostic & Estimate Builder:** guided selling with problem/solution carousels, flat-rate pricebook integration, “good-better-best” options; can present on tablet and capture customer signature on the spot.

- **Invoice & Payment:** invoice creation from estimate, apply discounts, take payment via card reader (Stripe/Worldpay), e-sign, print or email.
- **Time Tracking:** clock-in/out, drive time vs. job time, overtime calculations.
- **Inventory Management:** view truck stock, request transfer, scan barcodes for part usage, auto-deduct from inventory.
- **Offline Mode:** full job detail, forms, pricebook, and invoice capability; sync when connectivity resumes; conflict-free merge using CRDT.

2.4 Pricebook & Service Catalog

- **Pricebook Builder:** hierarchical catalog (trade → category → service → task), with material parts, labor hours, equipment, flat-rate prices; supports regional pricing and cost matrix.
- **Dynamic Pricing Engine:** price rules based on time-of-day, zone, membership discount, urgency fee.
- **Material & Equipment Library:** part numbers, supplier SKUs, cost, markup, taxability.
- **Proposal Templates:** branded PDF with cover page, photos, line items, financing options.

2.5 Invoicing, Payments & Accounting

- **Invoice Creation:** from job, estimate acceptance, or recurring membership; line items pull from pricebook or entered ad-hoc.
- **Payment Processing:** integrated with ServiceTitan Payments (powered by Stripe) or third-party gateways; accept credit/debit, ACH eCheck, cash, check.
- **Financing Options:** embedded consumer financing (GreenSky, Wells Fargo) application flow; decision in seconds.
- **AR & Collections:** aging reports, automated payment reminders via email/SMS, self-service payment portal for customers.
- **Accounting Sync:** native integrations to QuickBooks Online/Desktop, Sage Intacct; general ledger mapping, export journal entries, reconciliation.

2.6 Marketing Automation (TitanMarketing)

Personas: Marketing Manager, Owner.

- **Campaign Builder:** email, SMS, direct mail, triggered by lifecycle events (service anniversary, un-sold estimate follow-up, maintenance due, membership expiration).
- **Lead Management:** capture from web form, third-party lead aggregators (HomeAdvisor, Angi), phone call; auto-assign to CSR, track source ROI.
- **Online Booking (TitanAdvisor):** embeddable widget on contractor's website, real-time availability, capacity rules.
- **Reputation Management:** review request automations (Google, Facebook), review monitoring dashboard.

- **Data-Driven Segmentation:** dynamic lists based on recency, equipment age, service type, zip code.

2.7 Membership & Recurring Revenue

- **Plan Builder:** membership product with recurrence, auto-billing, included services, discounts.
- **Membership Dashboard:** contract status, next-service due, payment method expiry, up-sell opportunity prompts.
- **Auto-renewal & Billing:** integrated with ServiceTitan Payments for credit card/ACH recurring; handles failed payment retry logic and dunning management.

2.8 Inventory & Supply Chain

- **Warehouse Management:** multi-location inventory with min/max levels, reorder points, purchase order generation.
- **Truck Stock:** assign inventory to vehicles; technician can transfer parts, count reconciliation.
- **Vendor Catalog Sync:** electronic pricing updates from suppliers (Ferguson, Carrier, Goodman) via EDI or API.
- **Purchase Orders:** create PO from job or reorder; send to vendor, receive against PO, auto-update inventory value.

2.9 Reporting & Analytics

- **Standard Dashboards:** revenue summary, ticket counts, average ticket, technician sales, membership trends, callback rate, marketing source performance.
- **Custom Report Builder:** drag-and-drop dimensions (technician, trade, branch) and measures (revenue, hours, opportunities); exportable.
- **Alerts & KPI Targets:** set thresholds for daily booked revenue; push alert to manager if below target by 2 PM.
- **AI-Powered Insights:** anomaly detection on average ticket, seasonal recommendations, churn propensity.

2.10 Customer Communication & Portal

- **Two-way SMS:** CSRs and techs can text via platform with masked numbers; conversation logged.
- **Customer Portal (TitanCustomer):** view appointment details, approve/decline estimates, pay invoices, view service history, update contact info.
- **Automated Notifications:** appointment confirmation, "on my way," service summary with photos, follow-up survey, maintenance reminder.

2.11 Integration & Platform Extensibility

- **API Gateway (REST):** GraphQL and REST endpoints for jobs, customers, invoices, pricebook items, with OAuth2/API keys.
 - **Webhooks:** events on job status change, invoice payment, new lead.
 - **Marketplace Integrations:** 100+ pre-built connectors (Home Depot QuoteCenter, ServiceFinance, National Water, etc.).
 - **Data Import/Export:** bulk operations via CSV, automated ETL with Stitch/Fivetran to data warehouse.
 - **Telephony Integration:** deep CTI with CallRail, Dialpad, and Aircall; screen pop, click-to-dial, call recording linked to job.
-

3. Technical Architecture (C4 Model)

Level 1 — System Context

- **Contractor Users** → Web SPA (React), TitanPro Mobile (React Native/Kotlin Multiplatform), Integrations API
- **End Customers** → Customer Portal, SMS/Email Notifications
- **Third-party Systems** → Payment gateways, Accounting, Vendor Catalogs, Lead Aggregators, Telephony
- **ServiceTitan Platform** (multi-tenant SaaS)

Level 2 — Containers

[CDN] → [Web App (Angular/React)] [Mobile App] [Public API Gateway]

→ [Identity Service] (Auth0/Custom) → [Tenant Context Resolver]

→ [Microservices: JobSvc, CustSvc, InvSvc, PriceSvc, PaySvc, MktgSvc, InvtySvc, ReportSvc]

→ [Message Bus: Kafka/Pulsar] for async events (JobCompleted → InvoiceReady, PaymentReceived → AR update)

→ [Real-time Sync Engine] (WebSocket) for dispatch board live updates

→ [Database per bounded context: multi-tenant Postgres (row-level security) or separate schema]

→ [Search: Elasticsearch for customer/equipment global search]

→ [File Store: S3 for photos, signed docs, invoices]

→ [Data Warehouse: Snowflake] ← [CDC Stream] → [Analytics API for reports]

Level 3 — Key Components

- **Tenant-Aware Data Access:** a middleware decorates every query with `organization_id`; dynamic routing to db shard; partition map in Redis.
 - **Price Calculation Engine:** evaluates pricebook rules, membership discount, regional markup, tax; idempotent result cached per price set version.
 - **Dispatch Optimizer:** microservice solving VRP with constraints; uses GraphHopper/OSRM for distances; outputs reordered job sequence; result fed back via websocket.
 - **Payment Orchestration:** implements Stripe Payment Intents, records gateway-agnostic payment entity; handles refunds, partial capture, chargeback notification webhook.
 - **Form Engine:** JSON-schema-driven form renderer on mobile; supports conditional sections, photo capture, GPS stamp, barcode scan; results stored as structured JSON for reporting.
 - **Sync Engine (Mobile):** Couchbase Mobile/SQLite local store; sync gateway applies CRDT merge strategy on job status updates, form drafts; conflict resolver: last-writer-wins with server-side business rules overlay.
 - **Marketing Automation Engine:** campaign builder → segment evaluator (on cron/event trigger) → message dispatcher (SendGrid/Twilio/Lob) → action tracker (link clicks, open) → attribution to revenue.
-

4. Data Model (Simplified Core)

- **Organization** (tenant) → **Branch** → **Employee** (tech, CSR, manager)
 - **Customer** → **Location** (1:many) → **Equipment** (furnace, AC unit) with serial/model/age.
 - **Job** (work order) → **Task** (diagnose, repair, replace), **FormSubmission**, **TimeEntry**, **InventoryUsage**.
 - **Estimate** (proposal) → **EstimateItem** (product/service, good-better-best group) → **Invoice** (upon acceptance).
 - **Pricebook** → **PricebookItem** (material part or service) → **PriceModifier** (discount, surcharge).
 - **Payment** (transaction) → links to Invoice.
 - **MembershipContract** → recurring billing profile, service events.
 - **MarketingCampaign** → **CampaignRecipient** (lead/customer) → **Lead** → **Opportunity** (qualified lead to estimate).
-

5. User Flows & Experience Highlights

- **Call-to-Cash Flow:** Call in → CSR books job → tech receives push → on-site diagnoses, builds estimate from mobile, gets signature → one-tap convert to invoice, take payment → sync to accounting.
 - **Membership Lifecycle:** Tech offers membership during call → customer signs on tablet → contract activated, auto-bill set → recurring service reminders → auto-book at next due date.
 - **Real-time Dispatch Board:** drag job to new tech → capacities recompute, GPS ETA updated, customer SMS sent automatically.
 - **Technician Daily Workflow:** morning vehicle inspection form, clock-in, job navigation, on-site arrival geofence, dispatching parts request, time tracking, job completion review, clock-out.
-

6. Integration & API Design

- **Public API v2 (REST/JSON):**
 - `/customers`, `/locations`, `/jobs`, `/invoices`, `/payments`, `/pricebook/items`
 - Support `?$expand` for related entities, pagination with cursor.
 - Rate limits: tiered per app subscription.
 - **Webhook Events:** `job.created`, `job.status_changed`, `invoice.paid`, `membership.activated`, `lead.created`.
 - **Embedded Widgets:** book-online iframe with configurable theme, customer portal single sign-on with JWT.
 - **Partner App Platform:** third-party developers build apps that inject UI into TitanPro and Web; OAuth2 authorization code flow; scoped data access.
-

7. Infrastructure & Scalability

- **Cloud:** AWS, with multi-AZ deployment, cross-region DR for critical services.
- **Compute:** ECS Fargate for microservices, auto-scale on queue depth and CPU.
- **Database:** RDS PostgreSQL (sharded by `tenant_id` range), with pgbouncer pool; read replicas for reporting; Elasticsearch for full-text search.
- **Event Bus:** Managed Kafka (MSK) for job status change, payment, and marketing events; guarantees at-least-once delivery.
- **Cache:** Redis ElastiCache for session, price rules, feature flags (LaunchDarkly), dispatch optimization state.

- **CI/CD:** GitLab with Blue/Green deployments; feature flags for gradual rollout; canary testing on staging tenant.
 - **Disaster Recovery:** RPO < 1 min (continuous WAL shipping), RTO < 15 min; active-standby with manual promotion.
-

8. Security & Compliance

- **Authentication:** SAML/OIDC SSO, MFA enforced; API keys for integrations.
 - **Authorization:** RBAC with pre-defined roles (Admin, Manager, Tech, CSR) and custom permission sets; field-level masking for PII.
 - **PCI DSS Level 1:** card data never touches ServiceTitan servers; Stripe Elements/sdk tokenization.
 - **Data Encryption:** in-transit TLS 1.3, at-rest AES-256; key management via KMS.
 - **Audit Logging:** all record mutation logged with `changed_by`, `old_values`, `new_values` in immutable log table.
 - **SOC 2 Type II** certified, HIPAA-compliant (for patient-related info where applicable).
-

9. Observability & SRE

- **Metrics:** CloudWatch/Datadog dashboards for API latency, error rate, job throughput, payment processing success.
 - **SLIs:** API p95 < 500ms, Dispatch board update < 2s, Payment capture success > 99.5%, Mobile sync latency < 10s.
 - **Alerting:** on 5xx spike, payment gateway failure, dispatch optimizer queue backlog.
 - **Distributed Tracing:** W3C TraceContext propagated across all services; Jaeger visualization.
-

10. Roadmap & Future Orientation

- **AI-Driven Dispatch:** predictive time windows, tech-job suitability matching.
 - **Customer Self-Service Chatbot:** integrated with booking and invoice payment.
 - **Advanced Parts Marketplace:** inter-branch inventory sharing with blockchain-like ledger.
 - **Embedded Insurance:** offering equipment breakdown coverage at point-of-service.
-

This blueprint provides a holistic technical and feature overview of both EagleView and ServiceTitan, demonstrating how each platform's design, from pixel-level UI to infrastructure topology, directly serves the jobs-to-be-done of their respective users while anchoring to measurable business outcomes. The fusion of product, business, marketing, and architecture thinking is evident in every module, ensuring both platforms remain defensible, scalable, and deliver compelling user and economic value.

Strategic Blueprint: Implementation and Value of Next-Generation AI Architectures for Luminall's Ecosystem

Executive Summary: The Vision-Enabled Future of Luminall

This report presents a strategic blueprint for integrating state-of-the-art AI architectures into the Luminall ecosystem to capture and analyze unstructured visual data. By leveraging Vision Transformers (ViTs) and multimodal fusion techniques, Luminall can extend its natural language understanding (NLU) capabilities to real-world applications like property condition classification and sign detection. The analysis outlines a phased implementation roadmap, a robust benchmarking protocol, and a detailed ROI model, demonstrating how this expansion will not only create significant operational efficiencies but also unlock new, data-driven revenue streams and solidify Luminall's position as an end-to-end data intelligence platform.

1. State-of-the-Art AI Architectures for Computer Vision

A successful move into computer vision requires a careful selection of foundational AI architectures. The field has evolved significantly, offering a range of powerful models, each with distinct strengths and weaknesses. This analysis focuses on the most promising architectures for Luminall's specific use cases.

2. Convolutional Neural Networks (CNNs) and Hybrid Models

Convolutional Neural Networks (CNNs) have been the backbone of computer vision for years due to their efficiency in extracting features from images through convolutional filters.⁷ CNNs like AlexNet,

VGGNet, and ResNet have proven their effectiveness across a variety of tasks, with ResNet being particularly notable for enabling the training of very deep networks.⁷ For many applications, CNNs offer a balance of performance and computational efficiency, making them a reliable starting point. To enhance the performance of a single classifier, hybrid architectures have emerged as a powerful solution. A prime example is the CNN-XGBoost model. In this architecture, a CNN is used as a feature extractor, automatically processing an image to obtain a high-level representation, which is then converted into a vector.⁹ This vector serves as the input for a feature classifier, such as XGBoost, which is known for its superior accuracy and effectiveness in distinguishing complex features.⁹ This two-stage approach ensures that the CNN fully extracts relevant information, while the XGBoost model provides a robust and accurate classification, overcoming the limitations of a standalone classifier.⁹ The entire process can be further optimized by using techniques like Adaptive Particle Swarm Optimization (APSO) to fine-tune the hyperparameters of both the CNN and the XGBoost components, ensuring an optimal fusion of the two models.⁹

2.1 Vision Transformers (ViTs) and Multimodal Architectures

Vision Transformers (ViTs) represent a more recent and revolutionary approach to computer vision. Inspired by the Transformer architecture that dominates natural language processing (NLP), ViTs treat an image not as a grid of pixels to be convolved, but as a sequence of discrete patches or tokens.⁸ The model's key innovation is the multi-head self-attention mechanism, which allows it to model long-range dependencies across the entire image.¹¹ This is a significant advantage over CNNs, which have an inherently local receptive field.⁷ As a result, ViTs are remarkably robust to occlusions and other image disturbances, making them particularly well-suited for noisy, real-world data.¹³ However, this architectural flexibility comes with a trade-off: ViTs are less data-efficient than CNNs and typically require training on very large datasets to achieve state-of-the-art performance.⁸

For Luminall, the true strategic value of ViTs lies in their architectural synergy with the existing NLP-based core. Since the models that power Luminall's spreadsheet assistant are based on the same Transformer architecture as ViTs, they provide a natural bridge for multimodal learning.¹² This shared foundation is the key to seamlessly integrating visual and textual data. Multimodal models like CLIP (Contrastive Language-Image Pre-training) exemplify this fusion. CLIP uses a shared vision and language encoder to project both images and text into a common semantic space, where their relative proximity indicates their conceptual similarity.¹⁵ This capability unlocks powerful cross-modal applications like visual question answering (VQA), which allows a user to ask a natural language question about an image and receive a relevant response.¹⁷

The architectural decision to adopt Vision Transformers is a strategic move that goes beyond optimizing for a single computer vision task. By choosing a framework that is inherently compatible with its existing NLP stack, Luminall can create a unified "Transformer" ecosystem where visual and textual data can be processed and analyzed in a deeply integrated manner. This architectural commonality will streamline future development efforts, enable the creation of more sophisticated applications, and lay the groundwork for a future where intelligent agents, both visual and linguistic, can work in concert to automate complex, end-to-end workflows.¹⁹ The selection of this architecture is therefore not merely a technical choice but a strategic imperative that will enable Luminall to fulfill its promise as a versatile,

end-to-end data intelligence platform.

Table 1: Comparative Analysis of Core Computer Vision Architectures

Model Family	Key Strengths	Key Weaknesses	Suitability for Multimodal Fusion	Recommended Use Case
CNN	High data efficiency, established and optimized libraries	Limited receptive field, less robust to occlusions	Low	Initial Proof of Concept (PoC)
ViT	Models long-range dependencies, high robustness to occlusions	High data requirements, computationally expensive to train	High	Long-Term Strategic Architecture
Hybrid (CNN-XGBoost)	Combines robust feature extraction with high-accuracy classification	More complex two-stage pipeline, less flexible for fusion	Medium	High-Accuracy, Non-Real-Time Tasks

3. Benchmarking and Performance Evaluation Strategy

A successful implementation hinges on a clear and comprehensive evaluation strategy that goes beyond simple accuracy metrics. The benchmarking protocol must address both technical performance and real-world operational viability.

3.1. Technical Metrics

The choice of metrics depends on the specific task. For the Property Condition Classification task, which categorizes images into predefined classes, a suite of metrics is required to provide a complete picture of the model's effectiveness.²⁰

Accuracy measures the overall proportion of correct predictions, giving a general sense of performance.²⁰

Precision quantifies the proportion of positive predictions that were actually correct, which is critical for minimizing false positives, such as incorrectly flagging a property as damaged.²⁰

Recall, on the other hand, measures the proportion of all actual positive instances that were correctly identified, which is paramount for ensuring that critical defects are not missed.²⁰ The

F1-Score provides a balanced measure by combining precision and recall into a single value, which is particularly useful when dealing with uneven class distributions.²¹ A

Confusion Matrix offers a visual and detailed breakdown of where the model is succeeding and where it is making mistakes, identifying which classes are being misclassified.²⁰

For the Sign Detection task, which requires both object classification and localization, different metrics are needed.²⁰

Intersection over Union (IoU) is a geometric metric that measures the degree of overlap between the

model's predicted bounding box and the actual location of the object. A high IoU indicates precise localization.²⁰

Mean Average Precision (mAP) is the most common metric for object detection, as it provides a single, comprehensive measure of the model's overall performance by evaluating its precision and recall across all classes and at various IoU thresholds.²⁰

3.2. Operational Metrics

In addition to technical accuracy, the real-world utility of the models will be judged on operational metrics. **Inference Latency** measures the time it takes for a model to process an input and produce a prediction. This is a critical factor for real-time applications like live drone inspections, where quick decision-making is essential.²²

Throughput measures the number of predictions a model can generate per unit of time, which is crucial for large-scale, batch processing tasks such as analyzing an entire real estate portfolio of thousands of images.²² Finally,

Memory Usage and **Model Size** are key indicators for determining whether a model can be deployed on resource-constrained edge devices like smartphones or drones, where hardware and power are limited.²³

3.3. Benchmarking Protocol

The evaluation of these models requires a robust, repeatable protocol. This process begins with the creation of a high-quality, custom test set that accurately reflects real-world conditions.⁵ The quality of the dataset is paramount, as a model's performance is only as good as the data it is trained on.²⁴ To ensure the reliability of the results, a cross-validation strategy, such as k-fold cross-validation, should be employed to mitigate the risk of a single train-test split providing an overly optimistic view of performance.²⁰

For continuous monitoring and improvement, an AI benchmark dashboard should be established using tools like Google Data Studio or Tableau.²⁶ This dashboard will provide real-time insights into model performance, tracking key metrics over time to identify issues like performance degradation or data drift.²⁷ A standardized, code-based benchmarking script can be used to objectively measure and report on the inference latency and memory usage of different model architectures and configurations.²³

A successful deployment relies on a nuanced understanding of the trade-offs between a model's accuracy, speed, and size. For Luminall, a model with slightly lower accuracy that can be deployed on a drone for real-time, low-latency analysis may be more valuable than a more accurate model that requires expensive, high-latency cloud processing. The strategic decision is not about finding the single "best" model but about selecting the architecture that most effectively balances technical performance with the constraints and requirements of the target deployment scenario. This requires a holistic view of a model's capabilities and its operational costs, which this benchmarking strategy is designed to provide.

Table 2: Key Benchmarking Metrics by Task

Task	Primary Metrics	Explanation of Relevance	Deployment Metrics
Property Condition Classification	Accuracy, Precision, Recall, F1-Score	To assess the model's effectiveness in correctly categorizing images and balancing false positives vs. false negatives.	Latency, Model Size, Memory Usage
Sign Detection	Intersection over Union (IoU), Mean Average Precision (mAP)	To evaluate the model's ability to both correctly identify objects and precisely localize them with a bounding box.	Latency, Model Size, Memory Usage

4. Implementation Plan for Core Use Cases

A successful implementation requires a tailored plan for each use case, aligning the chosen architecture with the specific data and operational requirements of the task.

4.1. Property Condition Classification

For property condition classification, a fine-tuned Vision Transformer (ViT) model is the recommended long-term architectural choice. A pre-trained ViT can be fine-tuned on a custom dataset of property images to quickly adapt to the specific nuances of the task.²⁹ The ViT's inherent robustness to occlusions and perturbations makes it ideal for handling the imperfect, real-world images that are common in property inspections, which may feature poor lighting or partially obscured views.¹³ For an initial proof of concept, a hybrid CNN-XGBoost model could serve as a valuable alternative, as it may provide high accuracy on a smaller, more contained dataset.⁹

The data strategy for this task involves training the model on a dataset of images labeled with condition categories, such as "Excellent," "Needs Minor Repair," or "Severe Damage".²⁰ To enrich the model's output, it is valuable to fuse the visual data with other structured information, such as geospatial coordinates or property-specific metadata. The model's final output would be a structured data point—a classification or a condition score—that can be automatically populated into a Luminall spreadsheet. This integration allows a user to instantly assess the health of an entire real estate portfolio, enabling data-driven insights that were previously unavailable.¹

4.2. Sign Detection

The sign detection use case requires an object detection model that can accurately identify and localize objects within an image. A model built on a ViT backbone, such as those available through frameworks like Hugging Face, is an ideal choice for this task. This model will take an image as input

and produce a set of bounding box coordinates and a class label for each detected sign.²⁹

The next step in the pipeline is to use Optical Character Recognition (OCR) to extract the text from the signs identified by the object detection model.³³ The raw text from the OCR output is then processed and cleaned. This two-step process allows for the extraction of both the sign's location and its content. The final output is a clean, structured dataset with columns for "Sign Type," "Text," and "Coordinates," which can be seamlessly managed and analyzed within a Luminall spreadsheet. This automated process can be combined with other geospatial data, such as a drone's GPS logs, to create a comprehensive, geo-located inventory of all signs in a given area, which is invaluable for infrastructure management.¹

5. Integration with Luminall's Existing Ecosystem

The most critical and innovative aspect of this project is the integration of the new computer vision capabilities with Luminall's existing natural language core. Simply adding a new feature would fail to realize the platform's full potential; the goal is to create a seamless, cohesive experience.

5.1. Multimodal Fusion: Bridging Vision and Language

Luminall's success is built on its intuitive natural language interface for managing structured data.¹ To extend this value to unstructured visual data, the new vision models must be able to work in concert with the existing NLP stack. The solution lies in multimodal architectures that can align and fuse different data types, such as images and text.³⁵

Models like CLIP provide a powerful mechanism for this fusion.¹⁵ They use a shared encoder for both vision and language to project images and text into a common semantic space. The technical key to this fusion is the cross-attention mechanism, which enables the model to learn the relationships between text features and each pixel of an image, all while preserving the image's critical positional information.³⁵

This fusion redefines the user experience. Instead of a manual, multi-step workflow, a user can issue a single, intuitive natural language command within the Luminall interface. For example, a user could ask, "Find all properties with exposed wiring" or "Show me signs that say 'Road Closed.'" The system would use its NLP core to interpret the query and its integrated vision core to find and analyze the relevant images, even if "exposed wiring" was not a pre-defined classification category.

This fusion of vision and language is a strategic evolution of the Luminall user experience. It allows for a more fluid and human-like interaction with data, solidifying Luminall's brand promise as a platform that truly understands and processes information in a sophisticated way.¹ The benefit extends beyond automating a single task; it makes a new, complex workflow feel effortless and intuitive, thereby creating a significant competitive advantage and driving long-term value.

6. Deployment Strategy and Model Optimization

A strategic deployment plan is essential to move from a working prototype to a scalable, real-world solution. This involves a careful analysis of the trade-offs between cloud and edge deployment and a sophisticated approach to model optimization.

6.1. Cloud vs. Edge Deployment

The decision of where to run AI inference is a critical trade-off between power, latency, cost, and security.²⁴ Cloud deployment offers virtually unlimited computational power for training and fine-tuning large models. It is ideal for large-scale, non-real-time batch processing, such as analyzing a large portfolio of property images.²⁴

In contrast, running models on edge devices like drones or smartphones is crucial for real-time applications, such as a live sign detection system.²⁴ Edge AI provides several key advantages: it delivers immediate insights without relying on network connectivity, saves bandwidth by processing data locally, and enhances security by keeping sensitive information on the device.³⁷ The primary challenge with this approach is the limited computational power and memory of edge devices.²⁴

6.2. Model Compression and Efficiency

To deploy models effectively on resource-constrained edge devices, a comprehensive optimization strategy is essential.³⁸ The success of this effort depends on balancing accuracy with speed and efficiency.

Quantization is a technique that reduces the precision of a model's weights from a high-precision format (e.g., 32-bit floats) to a lower-precision format (e.g., 8-bit integers).⁴⁰ This can significantly reduce the model's memory footprint and speed up inference, which is critical for real-time and battery-powered applications.⁴⁰ Frameworks like TensorFlow Lite are specifically designed to enable this process.³⁸

Pruning focuses on removing redundant neurons or connections from a neural network that contribute little to its overall performance.³⁸ By "trimming the fat," pruning can reduce a model's size by a significant margin, which is crucial for devices with limited memory.³⁹

Knowledge Distillation is a more advanced technique that involves transferring the "knowledge" from a large, complex "teacher" model to a smaller, more efficient "student" model.⁴³ The student model is trained to mimic the outputs of the teacher, allowing it to achieve a similar level of performance with a fraction of the computational requirements.⁴³ This approach is particularly effective for edge deployment, as it enables the use of a more compact model without a significant loss in accuracy. Successful edge deployment is a multi-faceted engineering effort that combines architectural choices with a sophisticated toolchain of optimization techniques. The ability to orchestrate these different components—from selecting a lightweight model architecture to applying a sequence of quantization, pruning, and knowledge distillation—is essential for achieving the delicate balance of accuracy, speed, and size required for real-world application. This is a critical discussion point for any organization planning a large-scale AI deployment.

7. Strategic Value and ROI Analysis

The implementation of these AI architectures will deliver significant financial and strategic benefits for Luminall, moving its value proposition beyond simple cost savings to unlocking new, data-driven revenue streams.

7.1. Financial ROI

The most direct and immediate return on investment comes from automating tasks that are currently labor-intensive and costly. For example, AI-powered visual inspection systems can reduce the need for manual inspections by up to 90%, leading to substantial savings in labor, fuel, and vehicle costs for large-scale operations.⁴ In the real estate sector, automating property condition assessments eliminates the need for expensive on-site reviews, which can take days to complete.⁵

AI-driven inspection also enhances operational efficiency and reduces waste. Case studies show that automated visual inspection can increase defect detection rates by up to 90% compared to human inspection, reducing waste and preventing costly repairs by identifying issues early.⁴⁵ A well-trained AI system operates continuously without fatigue or distraction, providing consistent and reliable defect detection.⁴⁶ In the construction industry, drones equipped with AI have been shown to cut inspection time by 50% and reduce maintenance costs, which directly translates into a positive ROI by enabling faster data collection and more proactive maintenance.⁶ The return on investment can be calculated using a standard formula: $ROI (\%) = [(Financial\ Value - Project\ Cost) / Project\ Cost] \times 100$, where financial value is the sum of cost savings from labor, improved efficiency, and waste reduction.⁴⁵

7.2. Strategic Value Proposition

Beyond the quantifiable financial benefits, this initiative will provide significant strategic advantages. The ability to offer a platform that analyzes both structured and unstructured data will allow Luminall to stand out in a competitive market.⁵ This expansion creates new revenue opportunities by enabling data-driven services. By analyzing property images over time, for example, Luminall can track market fluctuations and asset health, providing new insights that can be monetized by investors and property managers.⁵

This project's ultimate value is the creation of a strategic intelligence platform from data that was previously unusable or too costly to collect at scale. The system moves from being a cost-reduction tool to a strategic asset that provides insights into market trends and asset health. This distinction is critical, as it transforms Luminall from a tactical tool into a strategic partner for its customers, cementing its brand as an innovative leader that helps organizations "do more with less".⁴⁹

8. Conclusion and Phased Recommendations

The strategic value of this initiative lies in its ability to transform Luminall from a specialized spreadsheet assistant into a multi-modal data intelligence platform. The technical analysis clearly indicates the superiority of Vision Transformers and multimodal fusion for this expansion, given their architectural compatibility with Luminall's existing NLP core. This will enable a seamless user experience where visual data is as easy to analyze as a spreadsheet.

To mitigate risk and ensure a smooth, sustainable transition, a phased implementation roadmap is recommended. This approach will allow the organization to manage the human and technical challenges of adopting new technology, as advised by business strategy experts.¹⁹

Table 3: Phased Implementation Roadmap

Phase	Key Objectives	Recommended Architecture	Key Milestones	Expected Outcomes
1: Proof of Concept (PoC)	Prove technical feasibility; establish performance baseline.	Hybrid CNN-XGBoost	Data collection, model training, initial benchmarking on a small dataset.	A clear baseline for accuracy, speed, and size; validated technical approach for a single use case.
2: Pilot and Optimization	Validate end-to-end value; optimize for real-world deployment.	Fine-tuned Vision Transformer (ViT) with multimodal fusion.	Real-world pilot with a key customer; application of quantization, pruning, and knowledge distillation.	End-to-end workflow from image to spreadsheet; quantifiable ROI metrics; optimized models for edge deployment.
3: Full-Scale Deployment	Achieve platform-wide integration; scale to new customer segments.	Integrated ViT-based multimodal platform with continuous monitoring.	Full market release; implementation of a live benchmark dashboard; expansion to additional use cases.	Maximum market impact; new revenue streams; enhanced brand perception as an end-to-end data intelligence platform.

By starting with a focused PoC and then moving to a pilot, Luminall can learn and adapt throughout the process, ensuring that the final solution is deeply integrated into its core business rather than being "bolted on" as a sidecar tool.¹⁹ This phased approach demonstrates a commitment to a measured, data-driven strategy, which is critical for earning trust and ensuring the long-term success of this

transformative project.

PROSPECT: Final Engineering & Solutions Architecture Document

1. Executive Summary & Business Alignment

Prospect is a unified real estate intelligence, renovation visualization, and investment strategy platform. It fuses predictive market data, generative AI (vision and text), and financial modeling into a single workspace for real estate investors, agents, and design-build firms.

1.1 Core Jobs-To-Be-Done (JTBD)

- **The Investor:** "Help me rapidly underwrite a batch of properties and simulate renovation ROIs so I can deploy capital with high confidence."
- **The Architect/Designer:** "Help me visualize structural and aesthetic interventions instantly to secure client buy-in and estimate scope costs."
- **The Agent:** "Help me generate compelling, data-backed marketing narratives and neighborhood analyses to close deals faster."

1.2 Unified KPIs (Product, Tech, Marketing)

- **Business:** Monthly Recurring Revenue (MRR), Customer Acquisition Cost (CAC), Net Dollar Retention (NDR).
- **Product:** Time-to-Analysis (TTA) < 30 seconds, Copilot engagement rate, Image Generation Acceptance Rate.
- **Technical (SLAs):** 99.99% Uptime, API P95 Latency < 200ms (excluding LLM calls), Zero data loss (RPO = 0).

2. Product & UX Architecture

Following Fitts's Law and cognitive load reduction principles, Prospect's interface is modular and contextual.

2.1 Core Modules

1. **Prospect Intelligence Batch (PIB):** A data-dense, grid-based interface for analyzing 1-50 properties simultaneously. Integrates real-time comparative market analysis (CMA).
2. **Renovation Visualizer Engine (RVE):** A visual-first canvas utilizing gemini-2.5-flash-image for layout modifications, coupled with modular step-by-step cost generation.
3. **Prospect Co-Pilot (Contextual AI):** An omnipresent, context-aware chat interface utilizing gemini-2.5-flash grounded in Google Search and local user state.

2.2 UX Engineering Principles

- **Optimistic UI & Silent Rehydration:** User inputs (property adds, chat messages) reflect instantly in the DOM while asynchronous Firebase/Firestore commits happen in the background.
- **Progressive Disclosure:** Advanced financial metrics and structural engineering warnings are hidden behind tooltips and expandable accordions to prevent initial cognitive overload.

3. System Architecture (C4 Model)

3.1 Context & Container Topology

Prospect utilizes a **Modular Monolith** pattern on the backend transitioning to **Microservices** for AI heavy lifting, adhering to Domain-Driven Design (DDD).

- **Client Application (SPA):** React 18, TailwindCSS, Firebase Web SDK.
- **API Gateway / BFF (Backend-for-Frontend):** Node.js/Express handling rate limiting, auth translation, and request routing.
- **Core Domain Services:**
 - *Property Context Service:* Manages property entities, CMA data fetching.
 - *Vision Service:* Orchestrates prompt engineering, payload construction, and Base64 image handling for the Google Generative AI APIs.
 - *Financial Engine:* Executes ROI, BRRRR, and amortized cost logic.
- **Data Persistence Layer:** Google Cloud Firestore (Primary OLTP), optimized for real-time document sync.

3.2 Bounded Contexts & Event Flow

Prospect relies on **Event-Driven Architecture** for asynchronous tasks to decouple the UX from slow LLM operations.

- `PropertyAnalyzedEvent` -> Triggers background CMA refresh.
- `RenovationImageRequestedEvent` -> Placed on Pub/Sub queue -> Vision Worker processes -> Updates Firestore -> React UI updates via `onSnapshot` listener.

4. Data Modeling & Persistence

Data structures are normalized for write efficiency but heavily denormalized into specific read-models (CQRS pattern) for sub-50ms UI rendering.

4.1 Core Entity-Relationship Schema (Firestore)

```
// Collection: users/{userId}/properties/{propertyId}
{
  "id": "uuid",
  "address": "123 Main St, TX",
  "baseline_metrics": { "beds": 3, "sqft": 1500, "purchase_price":
250000 },
  "renovation_scenarios": ["scenario_uuid_1", "scenario_uuid_2"],
  "created_at": "timestamp",
  "updated_at": "timestamp"
}
```

```
// Collection: users/{userId}/scenarios/{scenarioId}
{
  "id": "uuid",
  "property_id": "uuid",
  "industry_target": "Exterior Siding",
```

```

    "budget_tier": "Tier 2",
    "images": { "before_url": "gcs://...", "after_url": "gcs://..." },
    "financials": { "est_cost": 15000, "arv": 310000, "roi_percent": 300
  },
  "generated_scope": ["Demolish siding", "Install vapor barrier",
    "Install HardiePlank"]
}

```

4.2 Caching Strategy

- **Client-Side:** React Query / SWR for intelligent request deduping and stale-while-revalidate caching.
- **Edge:** CDN caching for static assets and compiled WASM bundles.

5. API & AI Integration Strategy

The platform heavily relies on the Gemini API suite. To manage cost (FinOps) and resilience, we employ strict integration patterns.

5.1 LLM Gateway Pattern

- **Idempotency & Retry Logic:** All external API calls (Gemini Text/Image/TTS) are wrapped in a resilient circuit breaker pattern with exponential backoff (1s, 2s, 4s, 8s).
- **Prompt Management:** System prompts are versioned and stored in a database, allowing product teams to run A/B multivariate tests on prompt efficacy without deploying code.
- **Payload Optimization:** Images are compressed and resized on the client-side via HTML5 Canvas before Base64 encoding to reduce network payload and API token consumption.

6. Infrastructure, Security, & FinOps

Prospect is built on a Cloud-Native, serverless-first infrastructure to map compute costs directly to active users (reducing idle overhead).

6.1 Cloud Topology (Terraform/CDK blueprint)

- **Hosting:** Firebase Hosting (Edge CDN).
- **Compute:** Google Cloud Run (Stateless containerized APIs, auto-scaling to 0).
- **Storage:** Cloud Storage (Blob storage for Before/After images with 30-day lifecycle policies for non-saved drafts).
- **Pub/Sub:** For decoupling AI generation tasks.

6.2 Security by Design (Zero Trust)

- **Auth:** Firebase Auth (JWT).
- **Data Sovereignty:** Firestore Security Rules strictly enforce multi-tenant isolation (`request.auth.uid == resource.data.userId`).
- **PII/Compliance:** Images uploaded containing people/faces are scrubbed or rejected

client-side before reaching the AI model to maintain privacy compliance.

6.3 FinOps Guardrails

- **Token Throttling:** Users are placed in tiers (Free, Pro, Enterprise). API calls to Gemini are intercepted and tracked in Redis. If a user exceeds their token budget, requests are gracefully degraded to cached responses or delayed queues.

7. Go-To-Market & Operational Readiness

The engineering architecture directly supports the marketing and growth strategy (Product-Led Growth).

7.1 Telemetry & Observability

- **Product Analytics (AARRR):** Custom event hooks embedded in React components track Feature_Adopted (e.g., generated a marketing email) and Conversion_Triggered (e.g., clicked "Upgrade for 4K Images").
- **Engineering Observability:** Datadog APM integration. Alerts triggered on:
 - Gemini API 429 (Too Many Requests) spikes.
 - P99 latency exceeding 5 seconds for text generation.

7.2 Deployment Pipeline (CI/CD)

- **Trunk-Based Development:** Feature branches merged to main trigger automated unit, integration, and contract tests.
- **Feature Flags (LaunchDarkly/Firebase Remote Config):** Enables Canary rollouts of new AI prompt structures to 5% of users to monitor failure rates before a global launch.
- **Rollback Strategy:** Immutable container tags in Cloud Run allow instant 1-click rollbacks if MTTR (Mean Time To Recovery) thresholds are breached.

End of Engineering Document. Approved for Prospect Launch Sequence.1. Analyze the Request:

User Roles (Provided):

Seasoned Software Engineer/Developer (Business-aligned product development, systems architecture, DDD, API design, agile methodologies, product-market fit).

Seasoned Product, Business, Marketing, and Solutions Architect (UX design, business process engineering, GTM strategy, cloud architecture, behavioral economics, cross-domain ecosystem mindset).

Task: "You and your team are tasked with creating the final engineering docs for prospect."

Constraint Checklist & Confidence Score:

1. Substantial, self-contained response? Yes.

2. File Generation? Yes.

3. Document format? Markdown (.md).

4. "" on own line? Yes.

5. Intro/Outro conversational? Yes.

Confidence Score: 5/5

Context: The user wants "final engineering docs for prospect". "Prospect" appears to be the name of the product or project. Based on the previous context (PropertyInsight AI Dashboard, real estate/renovation visualizer), it's highly likely that "Prospect" refers to this real estate/property intelligence platform, or a related CRM/pipeline tool for real estate professionals. I will frame the engineering documentation around a robust, scalable platform that aligns with the PropertyInsight AI dashboard features (Batch Analysis, Renovation Visualizer, AI Copilot) but elevates it to an enterprise-grade system architecture, fitting the highly advanced persona descriptions provided.

2. Determine Document Structure (Final Engineering Docs):

To satisfy both the Software Engineer and Solutions Architect personas, the documentation must cover business alignment, UX/Product, Architecture, Data, APIs, Infrastructure, and Go-To-Market/Operations.

Title: Prospect - Final Engineering & Solutions Architecture Document

Executive Summary & Business Alignment: Vision, JTBD, KPIs, Target Audience.

Product & UX Architecture: Core modules (Batch Analyzer, Renovation Visualizer, Co-Pilot), user flows, UI/UX principles.

System Architecture (C4 Model): Context, Containers, microservices breakdown.

Data Modeling & Persistence: Domain-Driven Design (DDD) contexts, schema overview, caching strategy.

API & Integration Strategy: API-first design, event-driven flows, third-party integrations (LLMs, MLS, GIS).

Infrastructure, Security & Cloud Topology: AWS/GCP blueprint, FinOps, Security (SOC2, Zero Trust).

Go-To-Market & Operational Readiness: Telemetry, deployment strategy (CI/CD), observability, scaling thresholds.

3. Drafting the Content (Persona Alignment):

Tone: Authoritative, highly technical yet business-focused, encyclopedic, structured.

Keywords to include: Domain-Driven Design, bounded contexts, Event Storming, CQRS, React, Node.js/Python, Vector Databases (for AI), Terraform, CI/CD, FinOps, JTBD, AARRR.

Application Context: "Prospect" (Real Estate Intelligence & Renovation Strategy Platform).

4. Refining the Markdown Structure:

Use clear headings, bullet points, and code blocks for JSON/Architectural definitions.

Ensure the document is comprehensive enough to act as a "final engineering doc."

5. Review against File Generation Rules:

Filepath: prospect_engineering_architecture.md

Language: markdown

Title: Prospect Platform Architecture

6. Execute Generation: (Drafting the document internally, ensuring all constraints are met). I have channeled the combined expertise of the systems architect, software engineer, and go-to-market solutions architect to produce the final, comprehensive engineering and product architecture documentation for Prospect.

This document aligns the underlying technical infrastructure, data models, and API boundaries directly with user psychology, business KPIs, and market scalability.